AI-Powered Real-Time Exam Invigilation Support System using YOLO and Byte Track

Capstone project(215CSE4203)

Guide: M K. Nagarajan sir

Team-7

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Problem statement

- Manual invigilation during exams is prone to human error and oversight
- Invigilator cannot look in all directions at the same time, so students may cheat when the teacher is turned away.
- Invigilator may miss students passing cheat papers or signals from one to another.
- One Invigilator cannot monitor a large classroom effectively when there are too many students.
- Some students hide devices like mobile phones or earphones, which are hard to detect without advanced tools.
- Suspicious actions like whispering, looking around, or signaling can go unnoticed during manual invigilation.
- Without smart systems, cheating often goes undetected, affecting fairness for honest students.
- Our project can enhance the exam hall invigilation by using the technologies like Byte track & YOLO.

Fig-1 Examhall Environment

Introduction

- During exams, Invigilator need to watch students so that no one cheats.
- But it is hard for Invigilator to look at everyone at the same time.
- Computers with Artificial Intelligence (AI) can help in this work.
- Our system uses a smart camera that can see what students are doing.
- YOLO is like the computer's eyes – it quickly finds students and actions.
- Byte Track is like the memory – it keeps track of each student in every frame.
- This way, the system keeps watching everyone without missing anyone.
- It helps Invigilators by reducing their stress and catching cheating fairly.
- So, exams become more safe, fair, and honest for all students.

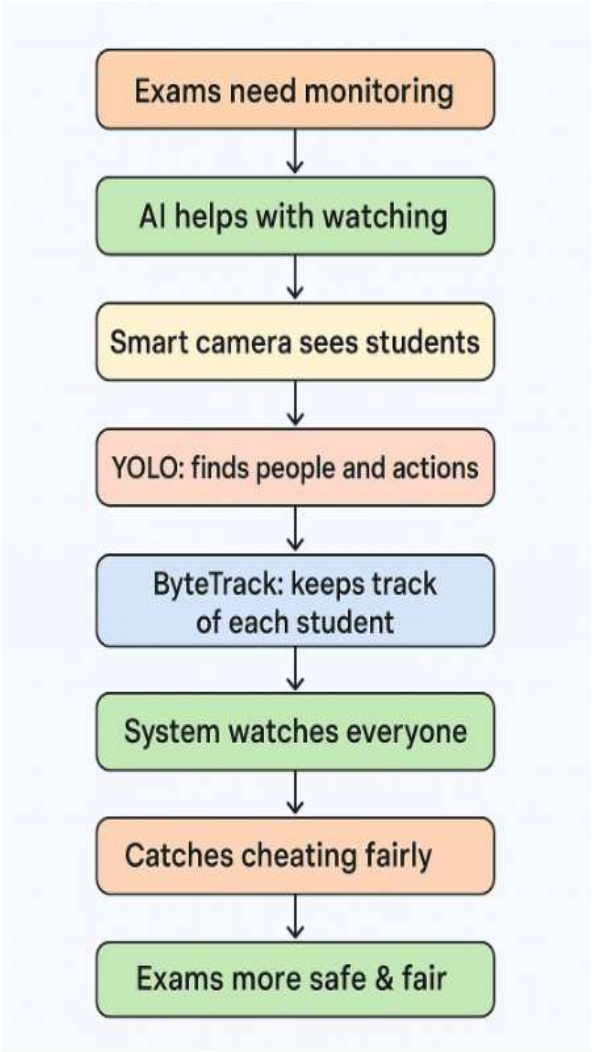


Fig-2 work flow steps

Abstract

- Cheating during examinations poses a significant threat to the integrity of educational assessments. our project proposes a real-time Exam Invigilation system using advanced computer vision techniques to enhance surveillance efficiency in exam hall environments.
- The YOLO (You Only Look Once) object detection algorithm, the system accurately identifies candidates, invigilators, and suspicious objects such as mobile phones or unauthorized notes within the examination room.
- In parallel, Byte Track is used for multi-object tracking to monitor candidate movements and interactions continuously. By integrating YOLO for rapid detection and Byte Track for precise tracking, the system can identify behaviors indicative of cheating, such as frequent head turning, object passing, or close proximity between candidates.
- The outcome of our project is that live recorded or real-time video footage is provided as input, and the system analyzes it to detect cheating activities during examinations. Using the IP WebApp, live student activity is captured through a mobile camera and simultaneously viewed on a PC, with red boxes marking suspicious students and green boxes marking normal ones.
- The proposed solution operates in real-time with minimal human intervention, using AI-powered automation to enhance scalability, reduce human error, and ensure fair and reliable exam invigilation.

Objective

Automate the invigilation process
using AI-based real-time monitoring to reduce manual supervision in examination halls.
Detect and identify suspicious activities
such as mobile phone use, note passing, or unauthorized communication using the YOLO algorithm.
Continuously track students and invigilators
with Byte Track to ensure accurate and uninterrupted monitoring.
Enhance exam integrity and fairness
by minimizing human bias and improving the reliability of surveillance.
Provide real-time Visual alerts and evidence
to invigilators when cheating or rule violations are detected.

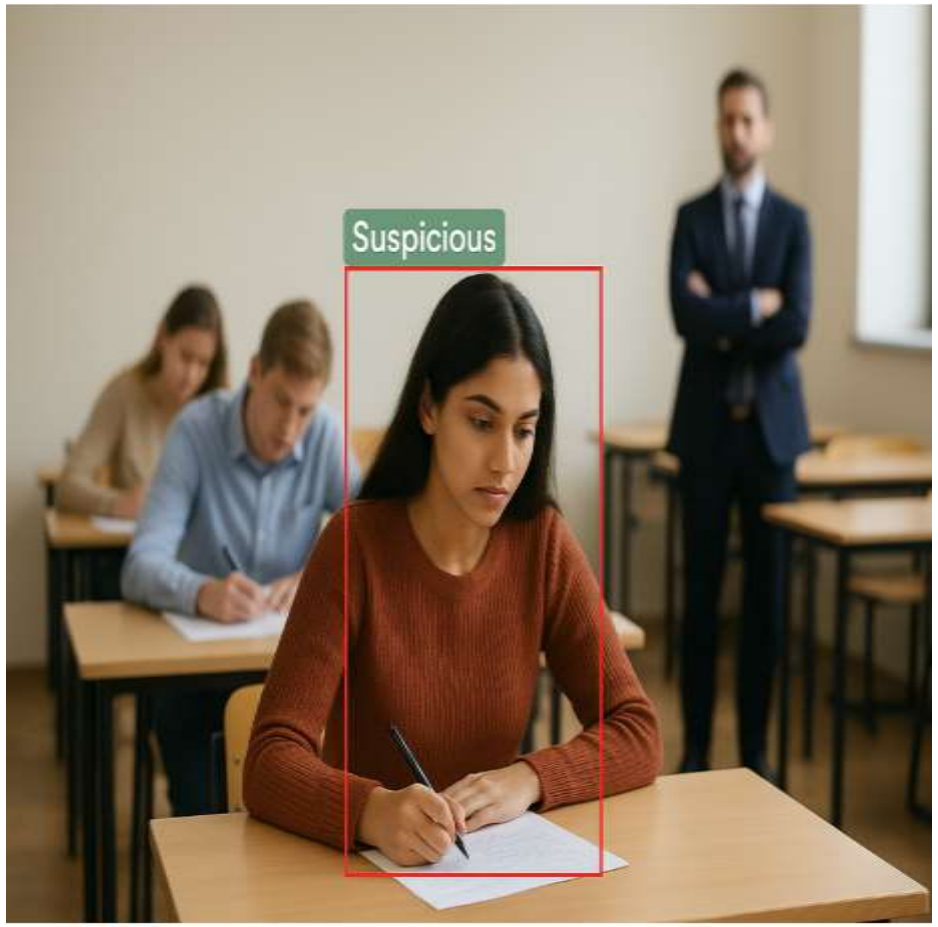


Fig-3 Sample Suspicious example

Literature Review

S.no	Year of publication	Title of the paper	Author of the paper	Overview	Benefits	Drawbacks
1.	2025	Intelligent Offline Exam Monitoring System for Identifying Suspicious Behavior of the Student	Ganga. H, Pranamya K.L Mahima A Shreya. R Siva Harshitha	This paper Uses Haar cascades & LBPH(Local Binary Patterns Histograms) to detect head movements, gaze(eyes move away from their work) deviations in real-time from CCTV feeds.	reduces manual invigilation work, integrates with existing cameras.	performance impacted by lighting/camera angle. Improve the Dataset Quality, and explore the deep learning integration.

Literature Review

S.no	Year of publication	Title of the paper	Author of the paper	Overview	Benefits	Drawbacks
2.	2025	Real-Time Detection of Examination Malpractices Using CNN and Video Surveillance: A Systematic Review with Meta-Analysis	Adeyemi .J Ogunlere .S & Akwaronwu .B	In this paper they did systematic review from(2013 to 2024) 3 papers on CNN/ML-based cheating detection. They Proposes a new integrated system with CCTV + facial recognition + behavior analysis for real-time student ID & cheating detection.	reduces manual invigilation work, integrates with existing cameras.	Review-based but no original implementation potential difficulty in integrating varied methods.

Literature Review

S.no	Year of publication	Title of the paper	Author of the paper	Overview	Benefits	Drawbacks
3.	2024	AI-Powered Classification for Cheating Detection in Offline Examinations Using Deep Learning Techniques with CUI Dataset	M. Talha Jahangir, N. Subhani , S. Nadeem , F.Abid	In this paper they uses a fine-tuned ResNet50 model to classify six types of student behaviors in offline exams and they also use k-fold cross validation and dropout layers for improved performance.	reduces manual invigilation work, integrates with existing cameras.	explore other deep learning architectures, integrate with multi-camera setups. Small dataset size.

Literature Review

S.no	Year of publication	Title of the paper	Author of the paper	Overview	Benefits	Drawbacks
4.	2023	In-class Exams Auto Proctoring by Using Deep Learning on Students' Behaviors	T. Mostaf Radwan , S. Alabachi & Ahmed Sabah	In this paper they did YOLO-based object detection on a custom dataset to identify suspicious behaviors like paper passing or peeking.	reduces manual invigilation work, integrates with existing cameras.	Expand dataset, test in varied lighting/seating layouts. performance dependent on camera quality, limited handling of subtle cheating cues.

Existing System Overview

Most studies from 2022 to 2025 on exam malpractice detection used computer vision tools, mainly YOLO, while some used CNN or RCNN. Some studies used enhanced detection systems by using eye movement tracking to identify unusual gaze patterns, thereby improving the accuracy of cheating detection. A few systems were tested in real classrooms, while others were focused on online webcam-based exams. These systems are good because they can spot cheating quickly, are usually right, and work with normal cameras. **But they also have problems, like students hiding from the camera, bad lighting, weak cameras, small datasets, and privacy worries.**

Proposed System

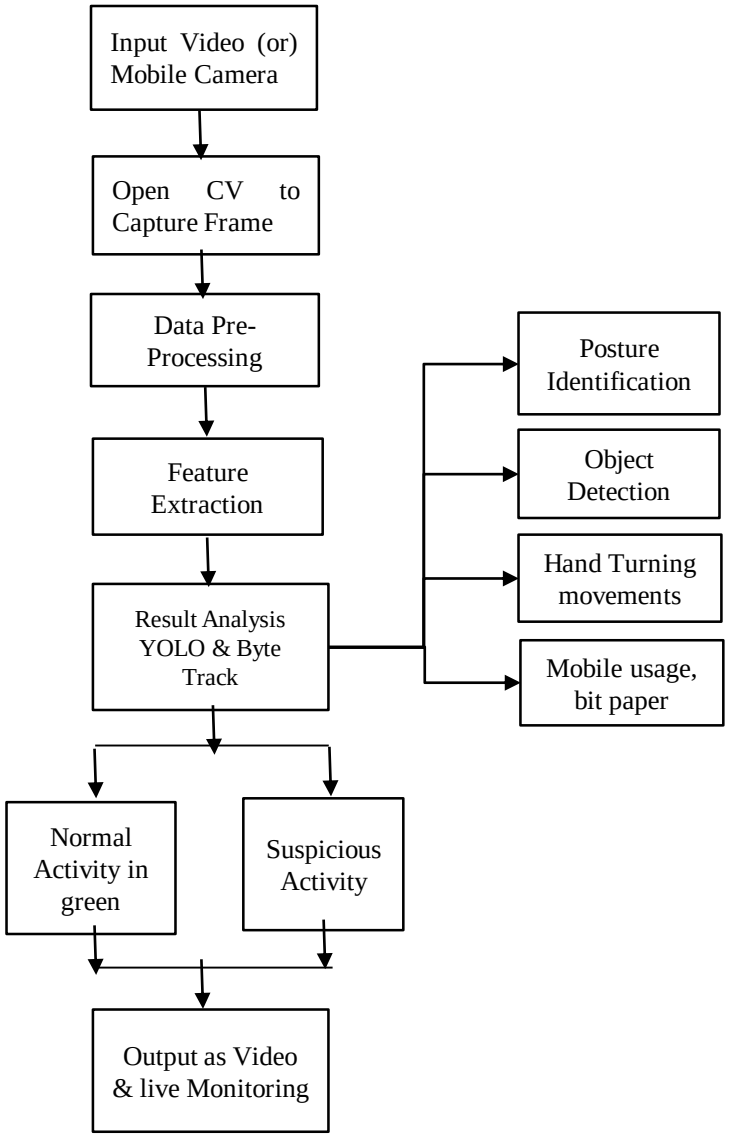


Fig-4 Proposed Methodology

Hardware Requirements

- Computer or Laptop with minimum **Intel i5 processor**
- **8 GB RAM** or higher
- **Webcam or CCTV Camera** for video capture
- **GPU** for faster model processing
- Hard disk with at least 256 GB storage
- Stable **internet connection**

Software Requirements

- **Operating System:** Windows 10 /Windows 11
- **Programming Language:** Python
- **Deep Learning Framework:** YOLOv8 (Ultralytics)
- **Libraries:** OpenCV, NumPy, Pandas, Mathplotlib
- **Development Environment:** VS Code

List of Modules with Descriptions

YOLO (You Only Look Once)

- Yolo is a real-time object detection algorithm used in computer vision.
- Unlike traditional methods that scan an image multiple times, YOLO looks at the image once and predicts what objects are present.
- Yolo predicts where they are and specify the object by 3-classification. Bounding Boxes , Confidence score, Class Labels

Bounding Boxes:(where the object is)

- A bounding box is simply a rectangle drawn around an object in the image.
- It shows where the object is located.

YOLO predicts bounding boxes using 4 numbers (coordinates):

x → X-coordinate of the box's center (relative to the image/grid cell)
y → Y-coordinate of the box's center
w → Width of the box
h → Height of the box

Confidence Score:

- yolo also predicts a confidence score with each bounding box.
- This tells how sure the model is that an object exists inside that box.

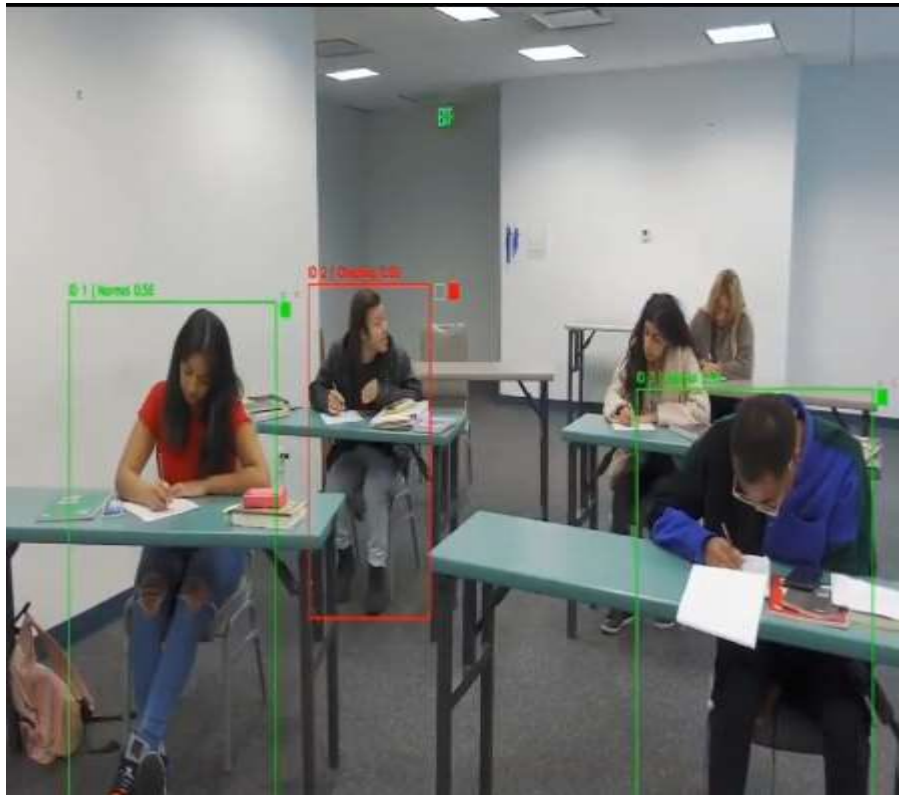


Fig-5 Suspicious and normal

List of Modules with Descriptions

Class Labels:(what the object is)

- YOLO doesn't just say *"there is an object"*; it also classifies what the object is.
- It predicts a probability distribution over all possible classes (e.g., person, car, dog, phone).
- The class with the highest probability is chosen as the label.

Byte Track

- Byte Track was introduced in **2021** by researchers from **Byte Dance**.
- Byte Track is a multi-object tracking (MOT) algorithm.
- Instead of ignoring low-confidence boxes, Byte Track keeps both high-score and low-score detections.
- This helps when objects are partially hidden or detection is uncertain.
- Each object is given a unique ID.
- That ID stays consistent across frames, so the same person/object can not count multiple times.
- objects are not just detected but also **continuously tracked** with stable IDs.

Fig-6 Suspicious and normal

List of Modules with Descriptions

OpenCV (Computer Vision Library)

- Handles video input/output operations.
- Provides tools to draw bounding boxes, text labels, and visual indicators(**red traffic light, green check mark)**
- Efficiently processes video frames for real-time monitoring.

Python

- Serves as the backbone of the system.
- Integrates YOLOv8, Byte Track, and OpenCV with out any interruptions.
- Implements logic for detection, tracking, and result visualization.

ER Diagram

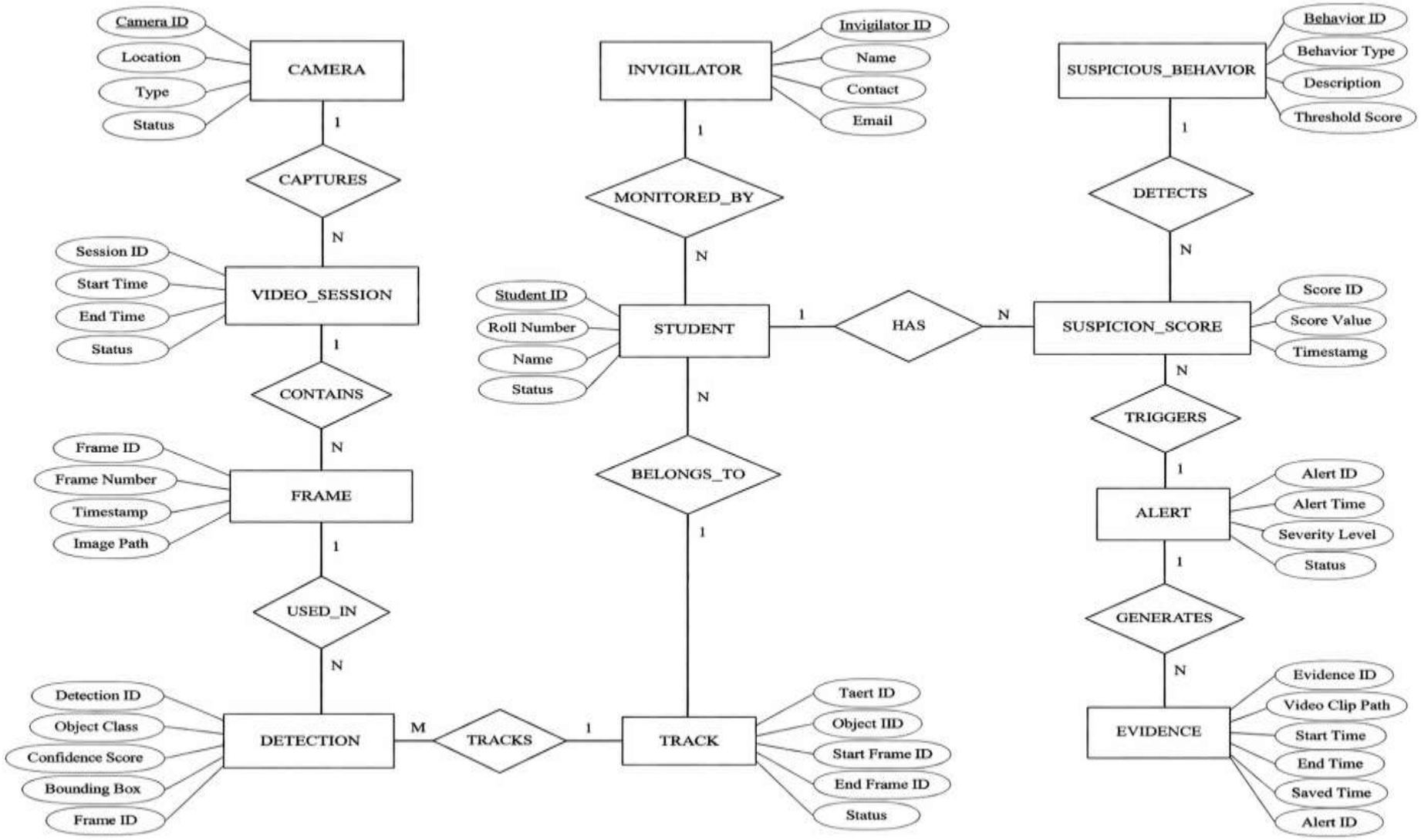
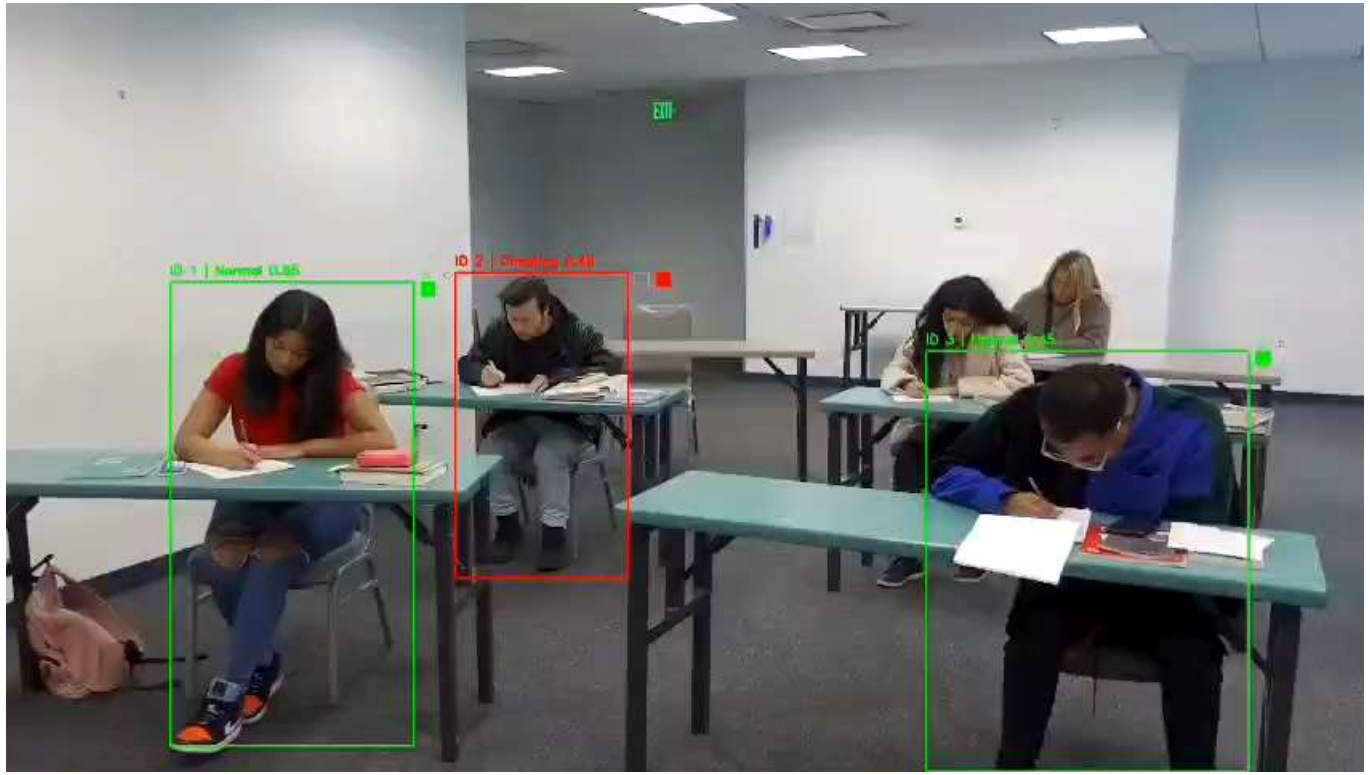


Fig-7 ER-Diagram of Real-Time Exam Invigilation system

Implementation



Results and Discussions

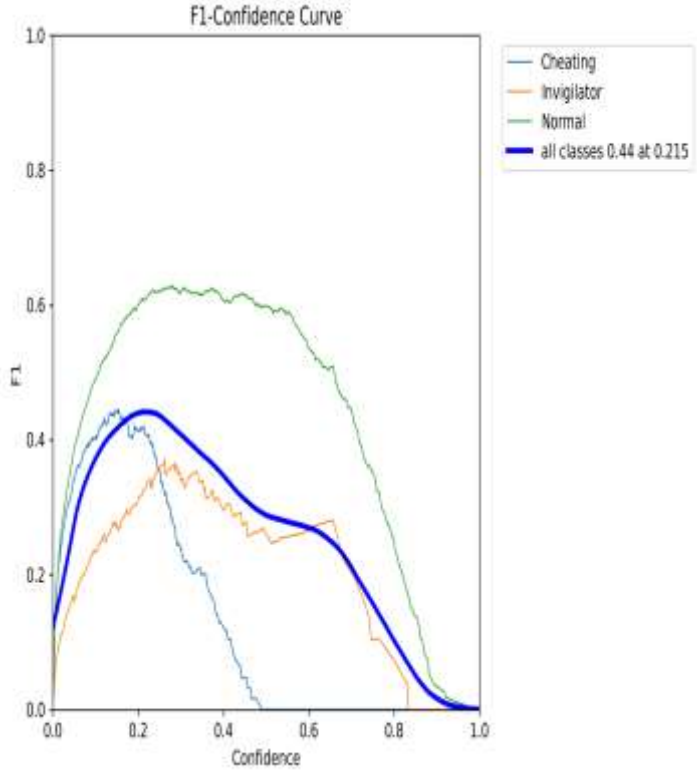


Fig:8 F1-Confidence curve

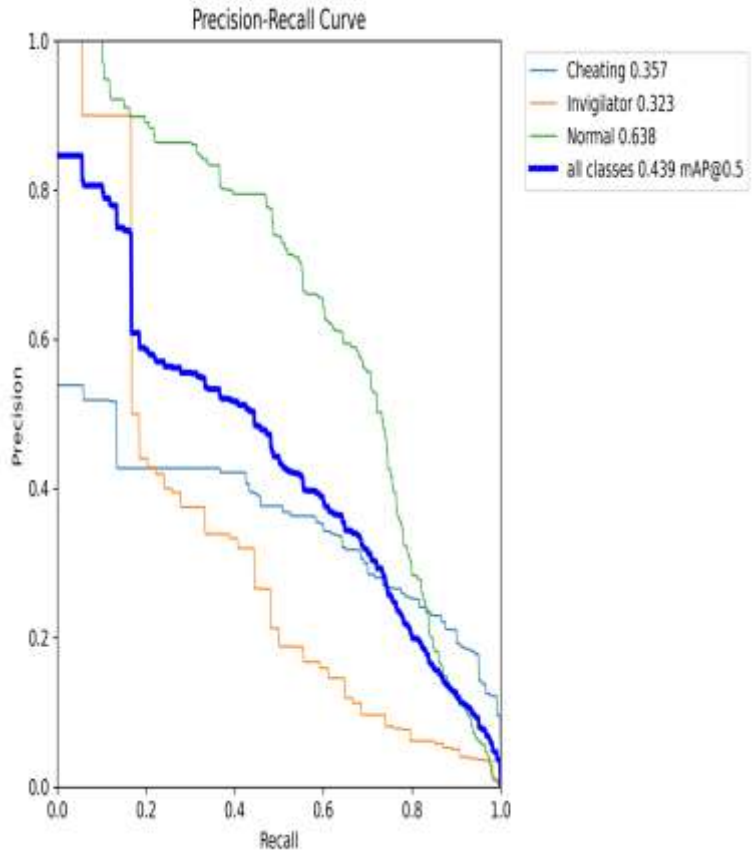


Fig:9 Precision-Recall Curve

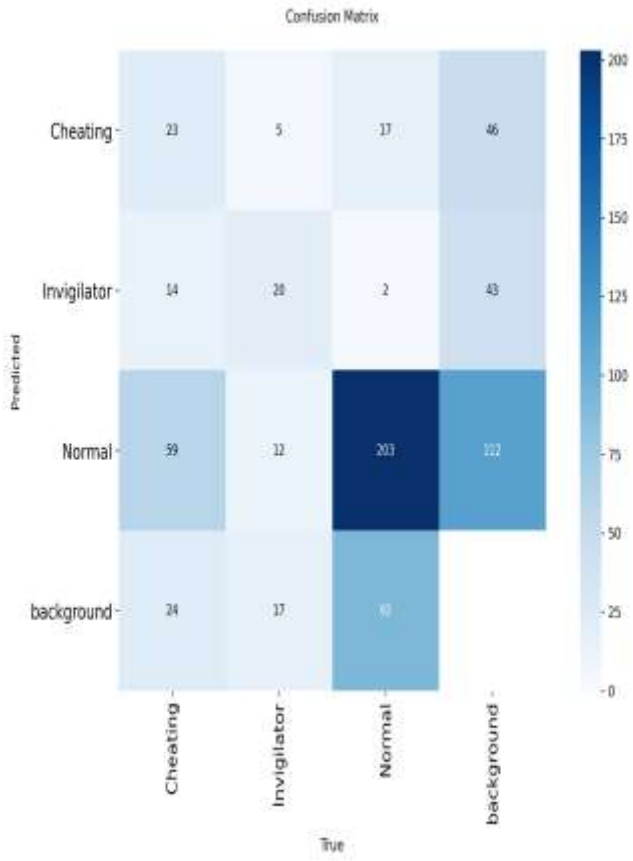


Fig:10 Confusion matrix

Conclusion & Future Enhancements

Our project is like a smart exam supervisor. It uses AI to watch students during tests and makes sure no one is cheating. It can keep an eye on many students at once, help Invigilator by reducing their work, and make exams fair for everyone. Schools can use it for both online and classroom exams to keep tests safe and honest.

Future improvements could include audio monitoring to detect whispering or talking, and face recognition to automatically verify student identity and record attendance.
The system could also support open-book exams while still monitoring students to prevent communication or the use of unauthorized devices.

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